

Assignment no. 6

For all exercises that require an LLM, use [Phi-4-mini](#) on your Azure ML instance.

Exercise 1 Prompting 1: Sentiment Analysis

For standard tasks, LLMs can also handle basic problems without training for a specific task. So let us revisit the IMDB dataset from Assignment 2 and use an LLM to perform sentiment analysis. To start with, evaluate the prediction quality using zero-shot prompting. To further improve performance, try to extend the prompt with examples (few-shot prompting). As LLMs are trained on vast amounts of data, it might be the case that a specific dataset was already used to train the model, which can skew the performance evaluation. Therefore, also investigate dataset contamination.

Tasks:

1. Come up with a usable prompt for the task.
2. Predict the review sentiment with zero-shot and few-shot prompting.
3. Evaluate the prediction performance for both prompting approaches and prepare the results for the exercise presentation.
4. Research dataset contamination for the presentation and also prepare potential solutions found in the literature.

Deliverables: A notebook containing your sentiment prediction, a comparison between zero-shot and few-shot prompting, and your research on dataset contamination.

Exercise 2 Prompting 2: Math Problems

To solve more complex problems, one can use chain-of-thought (CoT) prompting to produce better results. Let us compare zero-shot and CoT prompting on the [GSM8K math benchmark dataset](#). In this case, we only care about the final result, so design your prompt to make comparison with the final answer easy.

Steps:

1. Sample 100 examples from the GSM8K dataset to evaluate the prompting methods.
2. Come up with a usable prompt for the task.
3. Predict the final result using zero-shot and CoT prompting.
4. Evaluate the prediction performance for both prompting approaches and prepare the results for the exercise presentation.
5. Prepare to present a few examples of samples that were not solved correctly.

Deliverables:

- A notebook containing your GSM8K evaluation, a comparison between zero-shot and CoT prompting, and a presentation of the model's mistakes.

Exercise 3 Prompt Injection & Safety

As AI is used more broadly across the world, safety in LLMs is an increasingly important topic. While prompts can be used to restrict what LLMs are allowed to do, they can also be used to circumvent these restrictions. In this exercise, you will play both roles in the scenario. The goal of the attacker is to retrieve a secret code from the system prompt. Come up with defenses against leaking the code and then try to retrieve it using crafted prompts. Incrementally create protections and then attempt to bypass them.

Steps:

1. Create a system prompt that contains a code word.
2. Try to extract the code word by prompting the model.
3. Improve the system prompt to better defend against prompt injection.
4. Create several iterations of back-and-forth prompt hardening and attacks.

Deliverables:

- A notebook containing the prompt injection interactions.

When using Large Language Models (e.g., ChatGPT, Copilot, ...) to produce output that is submitted, also state the prompt and model used in the submission. Submissions that are not marked as such and are noticeably AI-generated will be graded with zero points!